1. Write the SOP and POS expressions:

   a. 2-input AND gate
   b. 2-input NOR gate
   c. 2-input NAND gate
   d. 2-input XOR gate
   e. 2-input XNOR gate

2. Write the SOP and POS expressions:

   a. 3-input OR gate
   b. 3-input XOR gate
   c. 3-input XNOR gate.

3. Prove that the SOP and POS expressions are equivalent.

   a. 2-input AND gate
   b. 2-input NOR gate
   c. 2-input NAND gate
   d. 2-input XOR gate
   e. 2-input XNOR gate

4. Prove the expressions using Boolean algebra.

   a. $(A+\overline{B})(\overline{A}+B)(\overline{A}+\overline{B}) = \overline{A}.\overline{B}$
   b. $(A+B)(\overline{A}+\overline{B}) = \overline{A}B + A\overline{B}$
   c. $\overline{A}.\overline{B} + \overline{A}B + A\overline{B} = \overline{A} + \overline{B}$
   d. $(A+B)(A+C) = A + BC$
   e. $\overline{A}B\overline{C} + AB\overline{C} + B\overline{C}D = B\overline{C}$
   f. $AB + A\overline{B} + CB + C\overline{B} = A + C$
   g. $AB + A\overline{B} + \overline{A}.\overline{B} + \overline{A}C + \overline{A}.\overline{C} = 1$
   h. $(A+B)BC + A = BC + A$
   i. $(A+B)\overline{B} + \overline{B} + BC = \overline{B} + C$
   j. $(A+\overline{B})(B+C)B = AB$
   k. $ABCD + AB\overline{C}D + ABC\overline{D} + A.\overline{B}.\overline{C}.D + AB\overline{C}.\overline{D} + A\overline{B}.\overline{C}.\overline{D} + A\overline{B}.C = A$
   l. $ABC + BC + \overline{B}C = C$
   m. $\overline{A}B\overline{C} + A\overline{B}C + AB\overline{C} + AB + A\overline{C} = A + B\overline{C}$

5. Prove the expressions using Boolean algebra and DeMorgan's theorems.

a. $\overline{\overline{A}.\overline{B}.(A+B)} = 1$

b. $\overline{A.\overline{B}} + \overline{A.(\overline{A}+C)} = \overline{A} + B + \overline{C}$

c. $\overline{\overline{AB}.(A+C)} + \overline{A}.B.\overline{(A+B+C)} = \overline{A} + B$

d. $\overline{\overline{\overline{ABCD}}} = (A+\overline{B})C + \overline{D}$

e. $\overline{(A+\overline{B})(\overline{A}+B)} = \overline{A}B + A\overline{B}$

f. $\overline{B}(A+\overline{C}) + \overline{\overline{A}BC + C(\overline{A}+B)} = \overline{B} + A.C$

g. $\overline{(\overline{A}+B+C+D)} + A.\overline{B}.\overline{C}.D = A.\overline{B}.\overline{C}$

h. $\overline{A}.\overline{B}.\overline{AB} = A + B$

i. $\overline{\overline{(AB+C)}(\overline{A}+\overline{B})\overline{C}} = AB + C$

j. $\overline{\overline{(A.\overline{B}.\overline{C}+BC)}}\,\overline{(AB)} = \overline{A} + B + C$

k. $\overline{\overline{(A+\overline{B})(C+D)}}\,\overline{(A+C)} = \overline{A}.\overline{B} + \overline{A}.\overline{C}$

6. Draw the expressions using only NAND gates.

a. $Y = AB + C$

b. $Y = ABC + D$

c. $Y = \overline{A}.\overline{B} + \overline{C}.D$

d. $Y = \overline{A}.\overline{B} + \overline{A}.B + A.B$

e. $Y = (A+B)(C+D)$

7. Draw the expressions using only NOR gates.

a. $Y = (A+B)C$

b. $Y = (A+B)(\overline{C}+\overline{D})$

c. $Y = A + BC$

d. $Y = A + BCD$

e. $Y = AB + C + D$